

APPLIED PHYSICS FOR CSE STREAM(BPHYS102/202)

OFFICIAL LINKS

 Website

<https://braintube.site>

 WhatsApp Channel (VTU Notes & Updates)

<https://wa.me/918884624741>

 YouTube Channel

<https://youtube.com/@braintube-v9w?si>

MODULE-1

LASER AND OPTICAL FIBERS

LASER PHYSICS

Pre-requisite: Properties of Light

Light is an electromagnetic wave characterized by:

- Wavelength λ
- Frequency ν
- Velocity c

Relationship:

$$c = \lambda\nu$$

Light interacts with matter by **absorption**, **emission**, and **scattering** of energy.

1. LASER – INTRODUCTION

LASER stands for:

Light Amplification by Stimulated Emission of Radiation

A laser is a device that produces an **intense, narrow, coherent beam of light** by stimulated emission.

2. CHARACTERISTIC PROPERTIES OF A LASER BEAM

Laser light differs from ordinary light due to the following properties:

1. Monochromaticity

- Laser emits light of **single wavelength**
- Extremely small spectral width

2. Coherence

- All waves are **in phase**
- Both **temporal** and **spatial coherence**

3. Directionality

- Laser beam travels in a **highly collimated** direction
- Very small divergence

4. High Intensity

- Energy is concentrated in a narrow beam
- Very high power density

3. INTERACTION OF RADIATION WITH MATTER

When radiation interacts with atoms, three processes occur:

1. Absorption

Atom absorbs photon and moves to higher energy state.

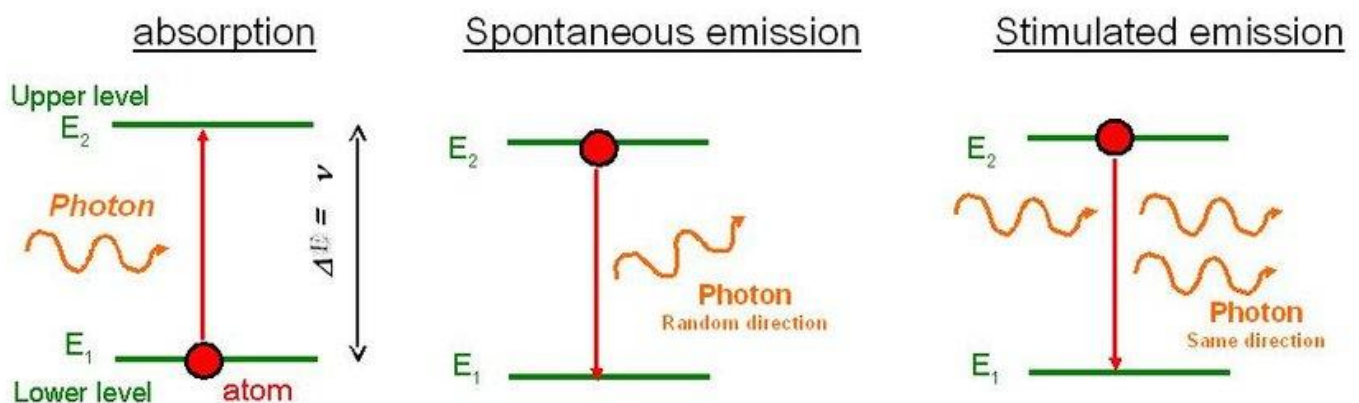
2. Spontaneous Emission

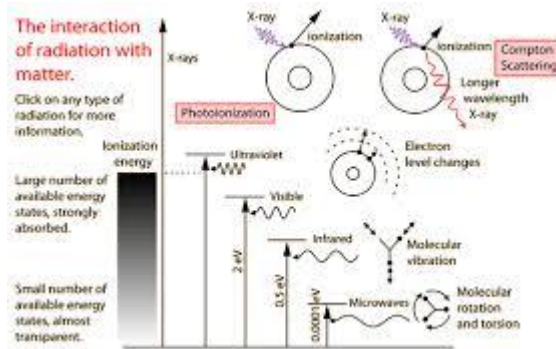
Excited atom returns to lower state by emitting photon randomly.

3. Stimulated Emission

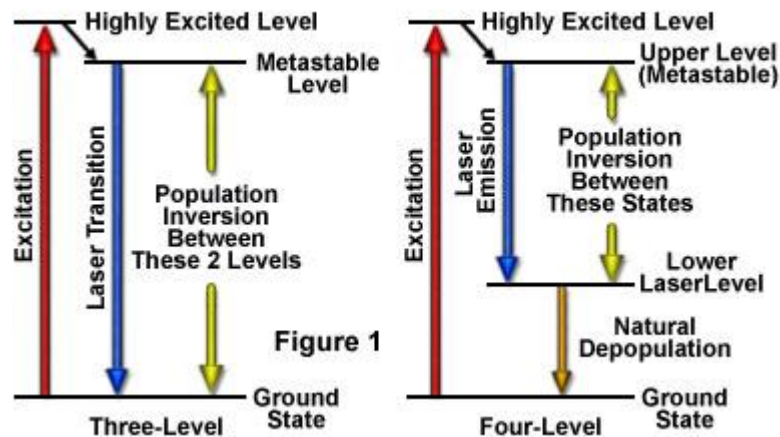
Incoming photon forces excited atom to emit a photon of **same energy, phase, and direction**.

Energy Level Interaction Diagram (REAL COLOR)





Three-Level and Four-Level Laser Energy Diagrams



4. EINSTEIN'S A AND B COEFFICIENTS

Einstein proposed three coefficients to explain radiation-matter interaction.

1. Einstein's A Coefficient

- Probability of **spontaneous emission**
- Independent of radiation density

2. Einstein's B_{12} Coefficient

- Probability of **absorption**

3. Einstein's B_{21} Coefficient

- Probability of **stimulated emission**

5. RELATION BETWEEN EINSTEIN COEFFICIENTS

From thermal equilibrium,

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

This leads to the expression for **energy density of radiatio**

6. EXPRESSION FOR ENERGY DENSITY (DERIVATION RESULT)

Energy density per unit frequency:

$$\rho(\nu) = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{e^{\frac{h\nu}{kT}} - 1}$$

This is **Planck's radiation formula**, derived using Einstein coefficients.

7. LASER ACTION

Laser action is possible only when:

- **Stimulated emission dominates absorption**

This condition is achieved by **population inversion**.

8. POPULATION INVERSION

Population inversion is a condition where:

$$N_2 > N_1$$

Meaning:

- More atoms in **excited state**
- Less atoms in **ground state**

This is opposite to thermal equilibrium.

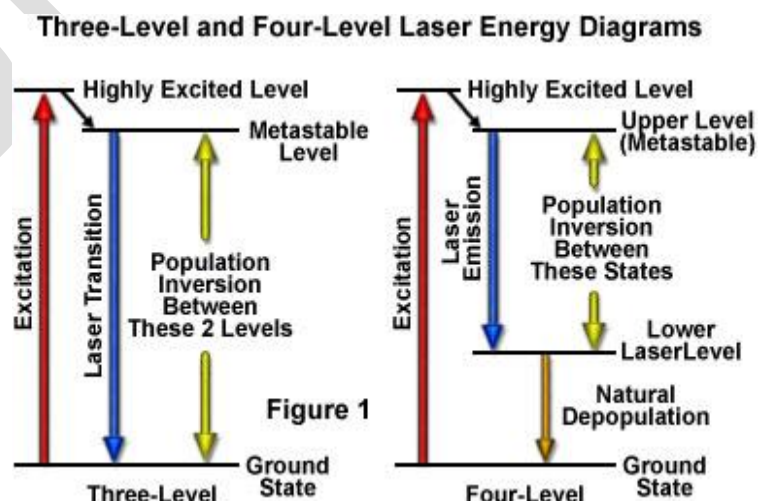
9. METASTABLE STATE

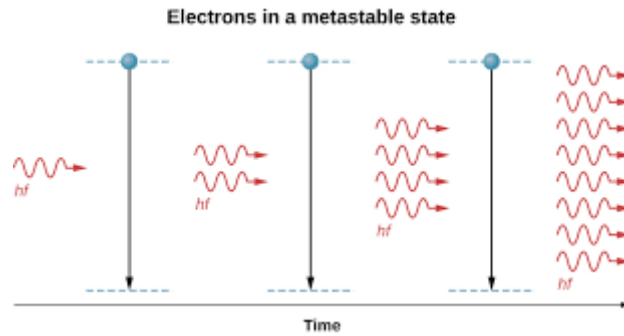
A **metastable state** is:

- An excited state with **long lifetime**
- Allows accumulation of atoms

It is essential to maintain population inversion.

Energy Level Diagram Showing Metastable State



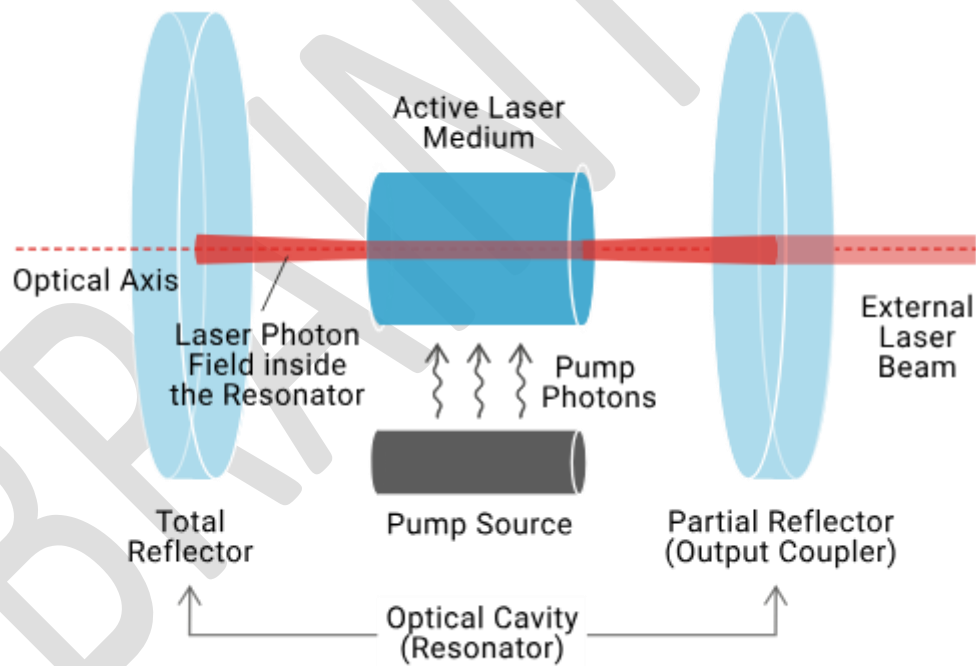


10. REQUISITES OF A LASER SYSTEM

A working laser requires

1. **Active medium**
2. **Pumping mechanism**
3. **Population inversion**
4. **Optical resonator (mirrors)**

Basic Laser System Diagram



SEMICONDUCTOR DIODE LASER

11. Semiconductor Diode Laser (SDL)

A **Semiconductor Diode Laser** is a compact laser source in which **stimulated emission** occurs within a **forward-biased p–n junction** of a semiconductor material (e.g., GaAs).

It directly converts **electrical energy into coherent optical energy**.

12. Construction of Semiconductor Diode Laser

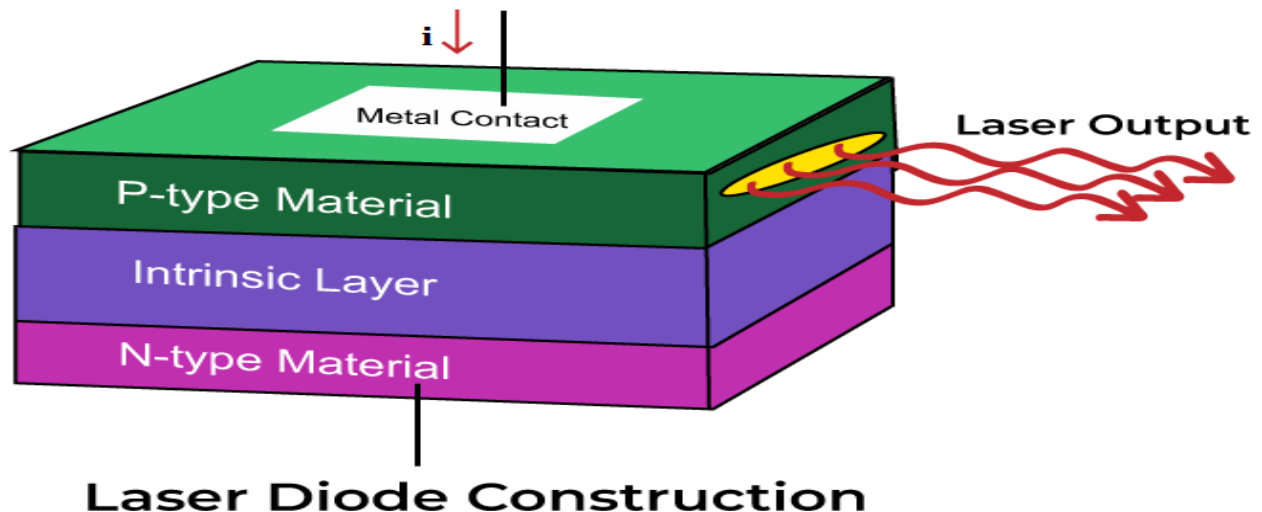
12.1 Materials Used

- Direct band-gap semiconductors such as **GaAs, InGaAs, AlGaAs**
- p-type and n-type regions form the **active junction**

12.2 Structural Parts

- **p-region** (acceptor doped)
- **n-region** (donor doped)
- **Active region** at the p-n junction
- **Ohmic contacts** for current injection
- **Cleaved end faces** act as mirrors (optical resonator)

Construction Diagram:



13. Working Principle of Semiconductor Diode Laser

13.1 Forward Biasing

- When the diode is **forward biased**, electrons from n-region and holes from p-region are injected into the **active region**.

13.2 Population Inversion

- At sufficiently high current, **population inversion** occurs in the active region.

13.3 Stimulated Emission

- Recombination of electrons and holes produces photons.
- Stimulated emission dominates, producing **coherent light**.

13.4 Optical Feedback

- Cleaved crystal surfaces reflect photons back and forth.
- Light gets amplified along the junction plane.

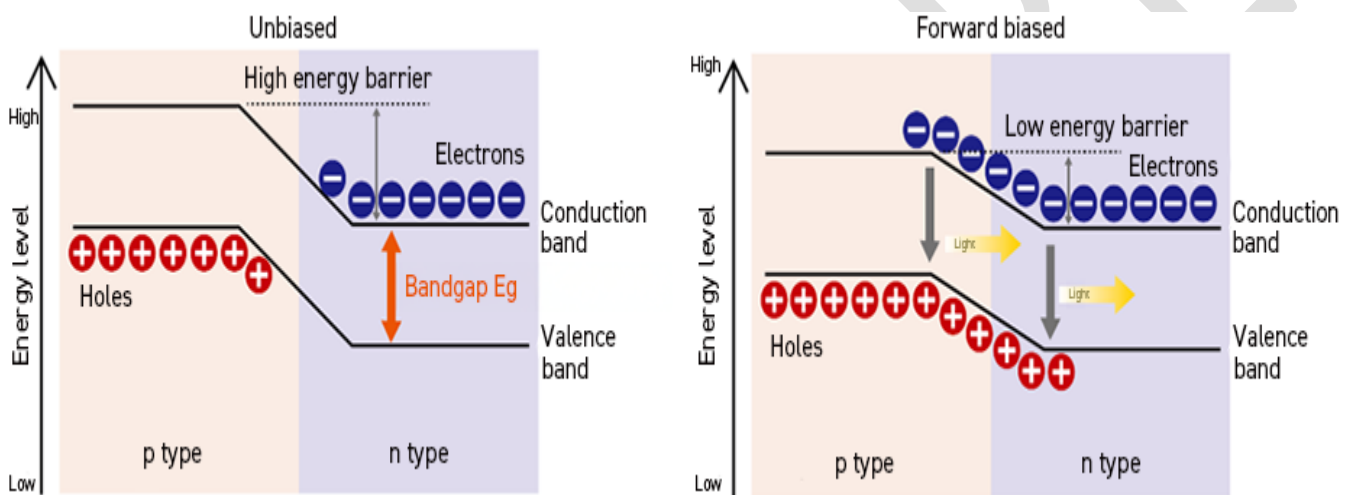
14. Energy Band Diagram of Semiconductor Diode Laser

The laser action in a semiconductor is explained using the **energy band model**.

Key Points

- **Conduction band (CB) and Valence band (VB)**
- Electron–hole recombination emits photons
- Direct band gap ensures efficient light emission

Energy Band Diagram



15. Threshold Current

The **threshold current** is the minimum current required to achieve population inversion and initiate laser action.

- Below threshold → LED-like emission
- Above threshold → Laser emission

Mathematically:

- Laser output power rises sharply beyond threshold current

16. Characteristics of Semiconductor Diode Laser

- High efficiency
- Low operating voltage
- Compact size
- Direct electrical pumping
- Fast response time

17. Advantages of Semiconductor Diode Laser

- Small and lightweight
- High reliability
- Low cost
- Long operational life
- Easy modulation

18. Disadvantages of Semiconductor Diode Laser

- Temperature sensitive
- Limited output power
- Requires cooling for stable operation

19. Applications of Semiconductor Diode Laser

19.1 Bar Code Scanner

- Uses focused laser beam to scan bar codes
- Fast and accurate data retrieval

19.2 Laser Printer

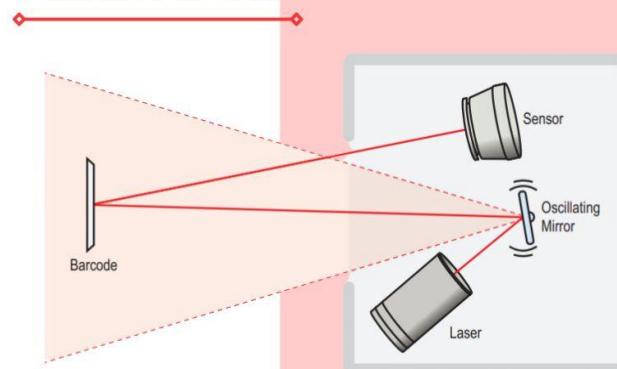
- Laser writes electrostatic image on drum
- Produces high-resolution text and graphics

19.3 Laser Cooling (Qualitative)

- Uses photon momentum to slow atomic motion
- Important in atomic physics and quantum experiments

Application Diagram

HOW DOES A LASER BARCODE SCANNER WORK?



20. Numerical Problems (Typical Examples)

Example

If the wavelength of a semiconductor laser is $\lambda = 850 \text{ nm}$, find the photon energy.

$$E = \frac{hc}{\lambda}$$
$$E = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{850 \times 10^{-9}}$$
$$E \approx 2.34 \times 10^{-19} \text{ J}$$

OPTICAL FIBER – BASICS & STRUCTURE

21. Introduction to Optical Fiber

An **optical fiber** is a thin, flexible dielectric waveguide that transmits light from one end to the other by the principle of **Total Internal Reflection (TIR)**.

Optical fibers are widely used in:

- High-speed communication
- Computer networks
- Medical and sensing applications

22. Principle of Optical Fiber

Total Internal Reflection (TIR)

Light traveling from a **denser medium** to a **rarer medium** undergoes **total internal reflection** when:

1. Angle of incidence i is greater than the critical angle i_c
2. Core refractive index $n_1 > n_2$ (cladding)

This ensures continuous propagation of light inside the fiber.

TIR Ray Diagram

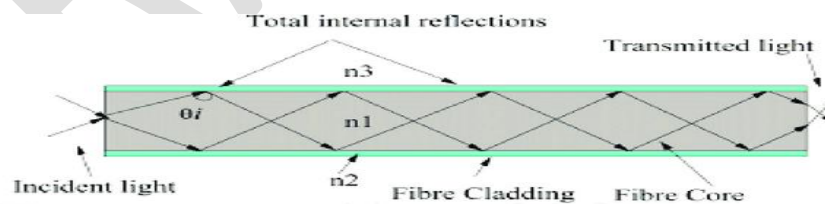
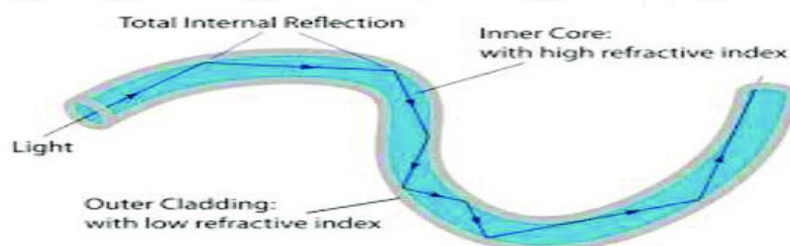


Figure 6.
Total internal reflection in an optical fiber.



23. Structure of an Optical Fiber

An optical fiber consists of three main parts:

23.1 Core

- Central region
- High refractive index n_1
- Carries light signals

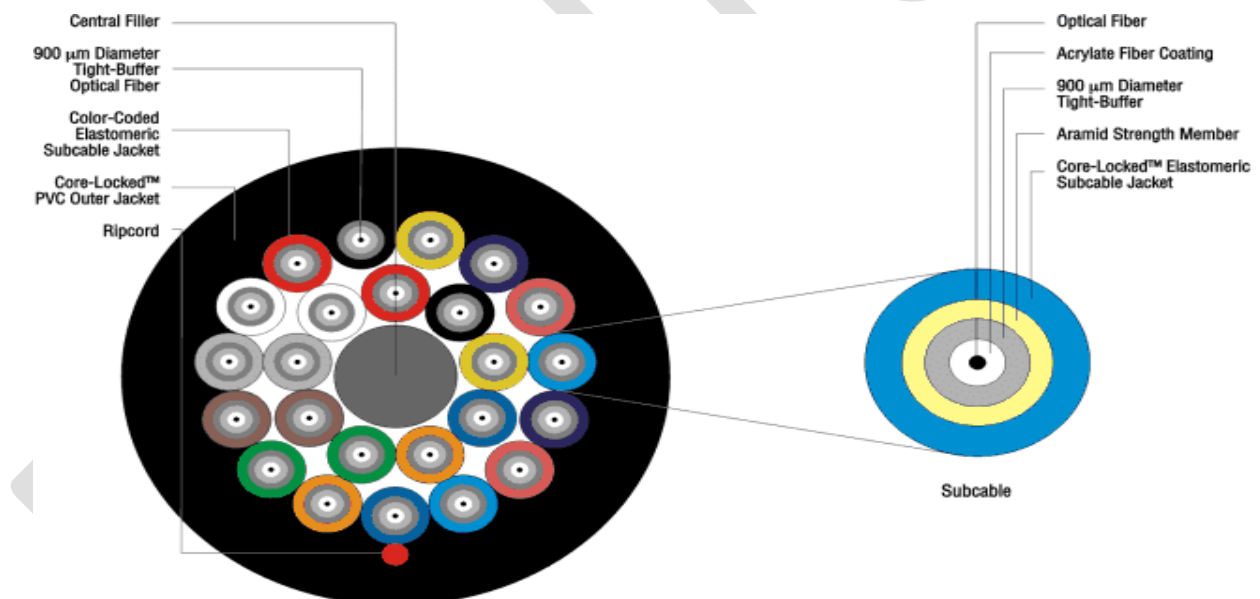
23.2 Cladding

- Surrounds the core
- Lower refractive index n_2
- Facilitates TIR

23.3 Protective Jacket

- Outer covering
- Mechanical strength and protection

Optical Fiber Structure Diagram



24. Propagation of Light in Optical Fiber

Light propagates through the core by repeated **total internal reflections** at the core–cladding interface.

Two possible ray paths:

- **Meridional rays** (pass through fiber axis)
- **Skew rays** (helical path, never cross axis)

25. Acceptance Angle

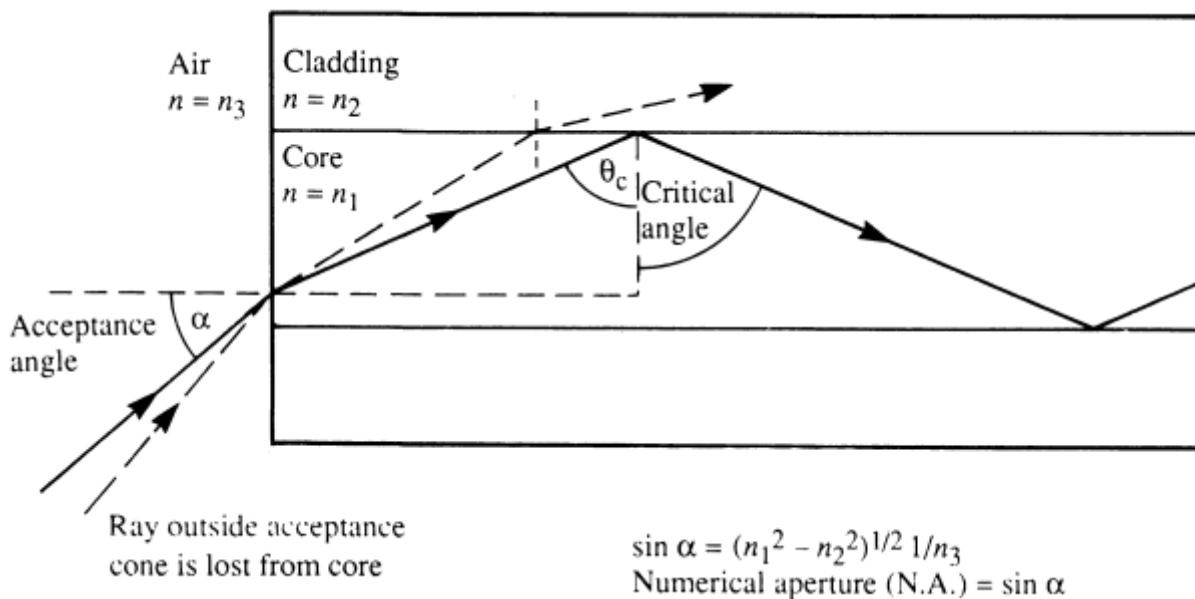
25.1 Definition

The **acceptance angle** (θ_a) is the **maximum angle** with the fiber axis at which light can enter the fiber and still undergo total internal reflection.

25.2 Acceptance Cone

- Incident rays within this cone propagate
- Rays outside the cone escape into cladding

Acceptance Cone Diagram



26. Numerical Aperture (NA)

26.1 Definition

Numerical Aperture (NA) is the **light-gathering ability** of the fiber.

$$\boxed{NA = \sin \theta_a}$$

27. Derivation of Expression for Numerical Aperture

Let:

- Core refractive index = n_1
- Cladding refractive index = n_2
- Refractive index of air = $n_0 = 1$

Using Snell's law and TIR condition:

$$\boxed{NA = \sqrt{n_1^2 - n_2^2}}$$

This formula determines the **maximum acceptance of light**.

28. Modes of Propagation

28.1 Single Mode Propagation

- Only one mode allowed
- Low dispersion
- Used for long-distance communication

28.2 Multimode Propagation

- Multiple modes propagate
- Higher dispersion
- Used for short distances

29. Refractive Index Profile (RI Profile)

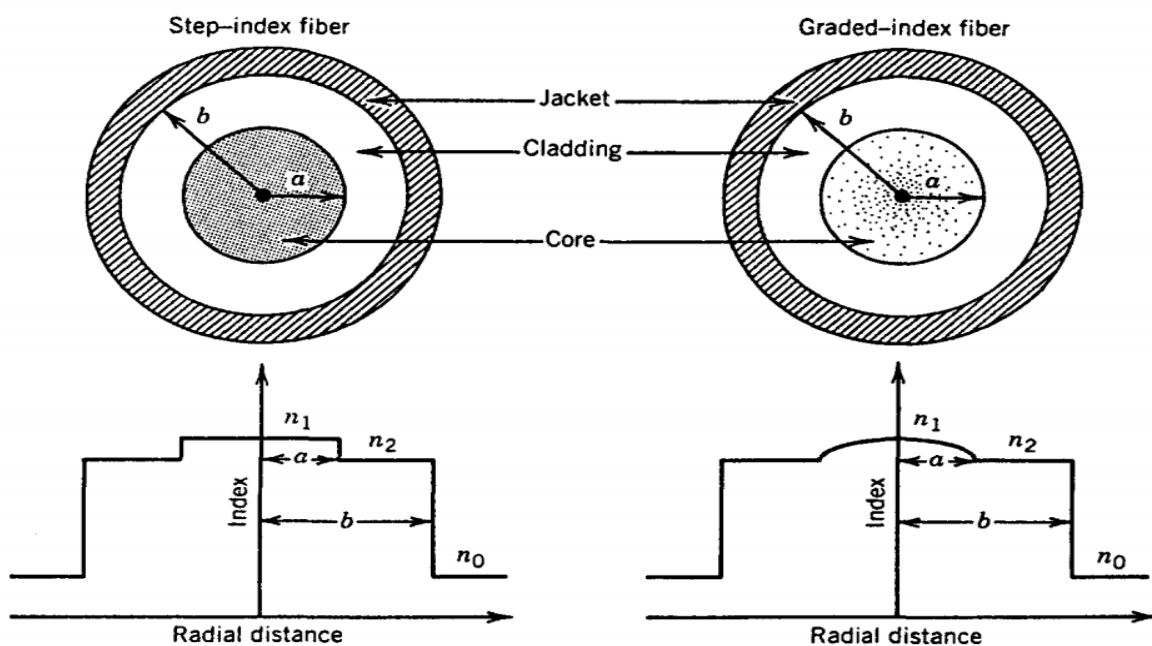
29.1 Step Index Fiber

- Abrupt change in refractive index
- Simple structure

29.2 Graded Index Fiber

- Gradual change in refractive index
- Reduced modal dispersion

RI Profile Diagrams



Cross section and refractive-index profile for step-index and graded-index fibers.

OPTICAL FIBER – CLASSIFICATION, LOSSES & APPLICATIONS

31. Classification of Optical Fibers

Optical fibers are classified based on modes of propagation and refractive index (RI) profile.

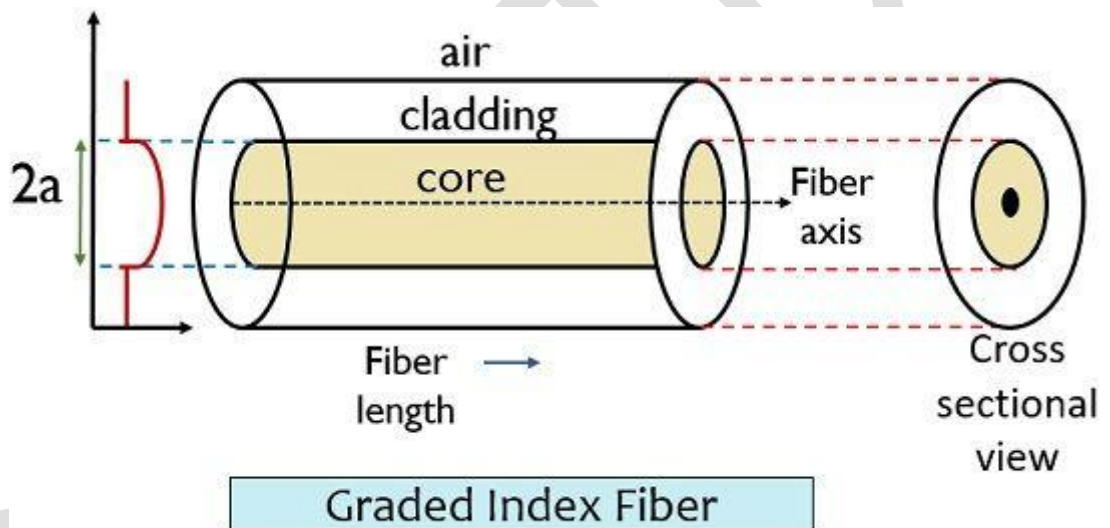
31.1 Based on Number of Modes

- Single-Mode Fiber (SMF)
Small core diameter, only one mode propagates, very low dispersion.
- Multimode Fiber (MMF)
Larger core diameter, multiple modes propagate, higher dispersion.

31.2 Based on Refractive Index Profile

- Step-Index Fiber
Abrupt change in refractive index at core-cladding boundary.
- Graded-Index Fiber
Gradual decrease of refractive index from center of core to cladding, reducing modal dispersion.

Fiber Types & RI Profiles



Circuit Globe

32. Attenuation in Optical Fibers

32.1 Definition

Attenuation is the **loss of optical power** as light propagates along the fiber.

It is measured in **dB/km**.

Attenuation coefficient:

$$\alpha = \frac{10}{L} \log_{10} \left(\frac{P_i}{P_o} \right) \text{ dB/km}$$

where

P_i = input power, P_o = output power, L = fiber length (km)

33. Mechanisms of Fiber Losses

33.1 Absorption Loss

- Due to impurities (OH^- ions, transition metals)
- Intrinsic absorption of glass material

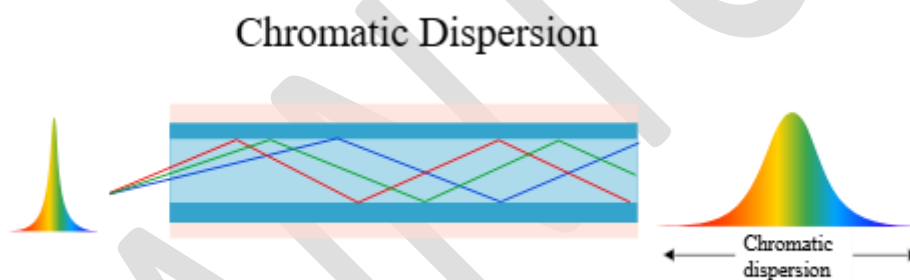
33.2 Scattering Loss

- **Rayleigh scattering** due to microscopic density fluctuations
- Dominant at shorter wavelengths

33.3 Bending Loss

- **Macrobending**: large bends cause radiation loss
- **Microbending**: small irregularities in fiber axis

Fiber Loss Mechanisms



34. Loss vs Wavelength Curve

- Minimum loss regions occur around **850 nm, 1310 nm, 1550 nm**
- **1550 nm window** offers least attenuation (~ 0.2 dB/km)

35. Applications of Optical Fibers

35.1 Fiber Optic Networking

- High-speed LAN and WAN connections
- Data centers and backbone networks
- Low signal loss and high bandwidth

35.2 Fiber Optic Communication

- Telephone and internet services
- Submarine communication cables

- Television signal transmission

Other Applications

- Medical endoscopy
- Fiber sensors
- Military and aerospace communication

36. Numerical Problems (Typical)

Example 1

An optical fiber of length **5 km** has input power **10 mW** and output power **5 mW**. Find attenuation in **dB/km**.

$$\alpha = \frac{10}{5} \log_{10} \left(\frac{10}{5} \right) = 2 \times 0.301 = 0.602 \text{ dB/km}$$

Example 2

If $n_1 = 1.48$ and $n_2 = 1.46$, find the numerical aperture.

$$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.48)^2 - (1.46)^2} \approx 0.24$$